CUBESAT B-DOT DETUMBLING AND HOHMANN TRANSFER ORBIT

Technical Project Report

Topics: Magnetic attitude detumbling, controller-gain sensitivity, magnetorquer saturation, and two-impulse orbital transfer.

Tools: Python • NumPy • SciPy • Matplotlib • GMAT

Prepared from the recorded simulation outputs and parameters provided for this project.

Abstract

This project studies two spacecraft-dynamics tasks relevant to a CubeSat mission. First, a simplified Python model simulates magnetic B-dot detumbling from an initial angular rate and evaluates how controller gain affects the time required to reduce the rate below 0.5°/s. The model also records commanded magnetic dipole magnitude and actuator saturation. Second, a two-impulse Hohmann transfer from a 500 km low Earth orbit to a target orbit with a 35,786 km apogee altitude is calculated analytically and checked in GMAT. The reported GMAT run reaches an apogee altitude of 35,786.52 km using a first-burn value of 2.3672 km/s. The results are preliminary and depend on the simplifications and setup assumptions described below.

1. Objectives

The objectives are to (1) model the reduction of spacecraft angular velocity using magnetic torque, (2) compare selected B-dot gains, (3) quantify magnetorquer command saturation, (4) calculate ideal Hohmann-transfer velocity increments and transfer time, and (5) compare the target apogee with the reported GMAT output.

2. B-Dot Detumbling Method

A magnetic torquer produces control torque according to $\tau = m \times B$, where m is the commanded magnetic dipole moment and B is the local magnetic field expressed in the body frame. The B-dot controller commands a dipole proportional to the negative body-frame magnetic-field derivative, with the command limited to the actuator's maximum magnitude. The spacecraft rotational equation is $I\dot{\omega} = \tau - \omega \times (I\omega)$, where I is the inertia tensor and ω is body angular velocity.

The implemented simulation uses a simplified Earth magnetic-dipole field along an idealized orbit and numerically integrates the attitude dynamics. The nominal gain is 50,000, the maximum magnetic dipole magnitude is 0.2 A·m², and the detumbling threshold is 0.5°/s. The nominal initial angular rate is [10°, 10°, 10°]/s; a second case uses [20°, 0°, 0°]/s.

2.1 Simulation Parameters

Parameter	Value
Initial orbit altitude	500 km
Orbital inclination in the simplified field model	97.4°
Inertia tensor	diag(0.042, 0.042, 0.0067) kg·m²
Maximum magnetic dipole moment	0.2 A·m²
Nominal controller gain	50,000
Detumbling threshold	0.5°/s

3. Detumbling Results

For the nominal initial angular rate [10°, 10°, 10°]/s, the recorded time to fall below 0.5°/s is 57.91 minutes. For the alternate initial case [20°, 0°, 0°]/s, the recorded threshold time is 85.47 minutes. These values are simulation outputs for the stated model and should not be interpreted as validated flight performance.

3.1 Gain Comparison

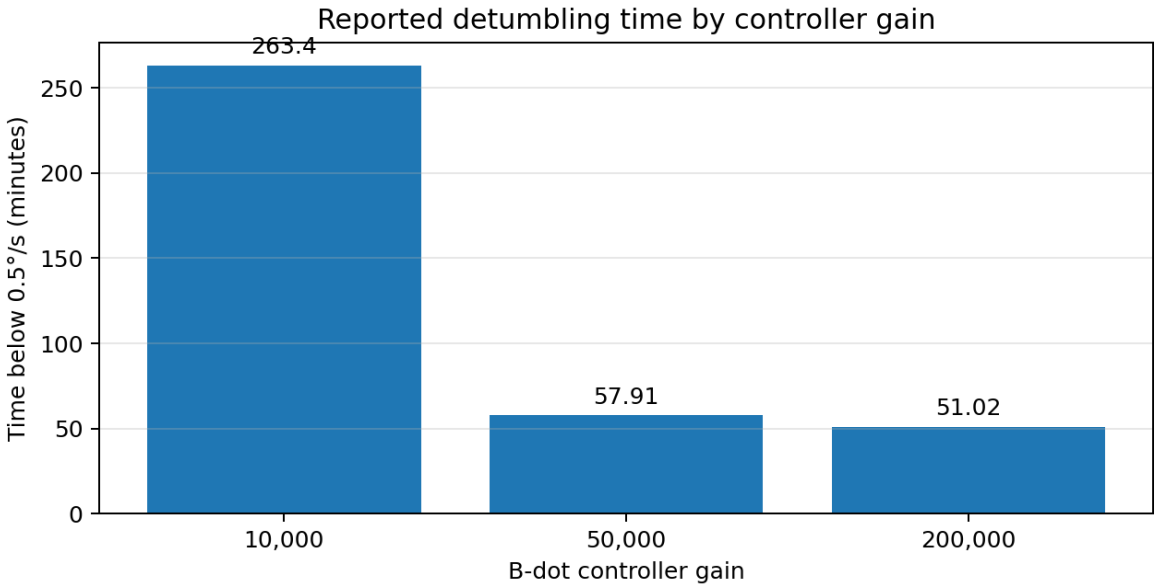


Figure 1. Recreated bar chart using the reported gain-comparison values. The original saved plot was not available to this report-generation session, so this figure is reconstructed from the recorded numerical results.

Gain	Time below 0.5°/s
10,000	263.4 min
50,000	57.91 min
200,000	51.02 min

The tested higher gains produced shorter detumbling times in this simulation, although the improvement from 50,000 to 200,000 was relatively small. Gain selection should also account for actuator saturation, measurement noise, sampling, and robustness, which are not fully represented in this simplified model.

3.2 Magnetorquer Saturation

The dipole analysis reports a maximum commanded magnitude of 0.2000 A·m² and approximately 13.50% of evaluated samples at saturation. This indicates that the controller request reached the configured actuator limit for a portion of the simulation. The saved detailed plot is identified as *results/dipole_magnitude.png*; it was not directly available for embedding in this report-generation session.

4. Hohmann Transfer Analysis

A Hohmann transfer is an ideal two-impulse maneuver between coplanar circular orbits. For Earth gravitational parameter μ, initial orbital radius r■ and target orbital radius r■, the transfer ellipse semi-major axis is a = (r■ + r■)/2. The initial and final circular speeds are v■ = √(μ/r■) and v■ = √(μ/r■). The transfer speeds at perigee and apogee follow the vis-viva equation v = √[μ(2/r – 1/a)].

The first ideal burn raises the spacecraft from the initial circular orbit to the transfer ellipse; the second ideal burn circularizes at the target radius. The ideal transfer time is half the transfer ellipse period: t = π√(a³/μ).

4.1 Analytical Values

Quantity	Value
Earth gravitational parameter	398600.4418 km³/s²
Initial altitude / radius	500 km / 6,878 km
Target apogee altitude / radius	35,786 km / 42,164 km
Transfer semi-major axis	24,521 km
Initial circular speed	7.6127 km/s
Transfer perigee speed	9.9825 km/s
Analytical first-burn magnitude Δv■	2.3698 km/s
Transfer apogee speed	1.6284 km/s
Final circular speed	3.0747 km/s
Analytical second-burn magnitude Δv■	1.4463 km/s
Total ideal Δv	3.8161 km/s
Ideal transfer time	Approximately 5.31 h

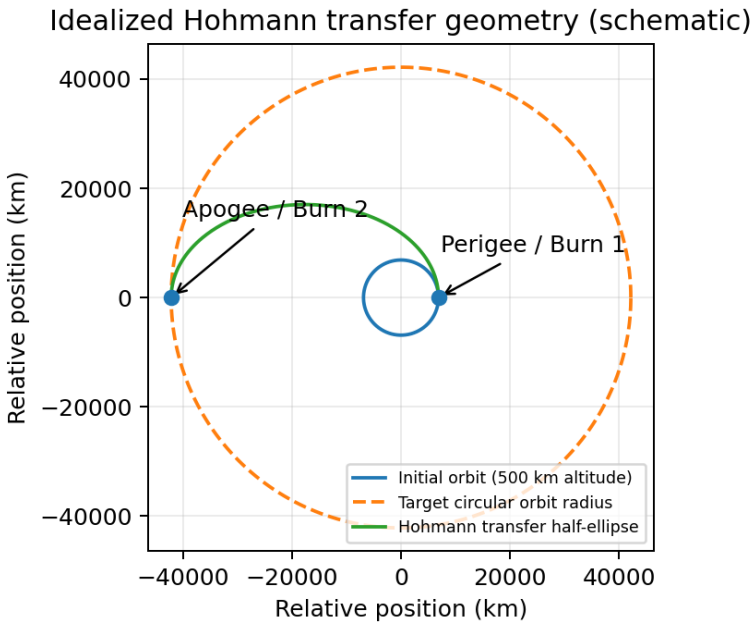


Figure 2. Schematic geometry of the idealized Hohmann transfer. This diagram illustrates the orbital geometry and is not a reconstruction of the GMAT trajectory.

5. GMAT Simulation and Comparison

The GMAT mission was configured with a spacecraft in a low Earth orbit, an impulsive maneuver using a VNB local frame, and propagation to the transfer orbit's apogee. The recorded final first-burn value is 2.3672 km/s. The reported apogee altitude is 35,786.5186 km, approximately 0.5186 km above the target altitude of 35,786 km. This is a relative altitude difference of about 0.00145%.

The configured GMAT burn is not exactly the same as the ideal analytical ∆v■ of 2.3698 km/s. Differences can result from the actual initial orbital elements, residual eccentricity, maneuver frame and orientation, propagation setup, and the exact state at which the maneuver is applied. The apogee match shows that the reported run reached close to the requested altitude; it does not by itself validate every aspect of an ideal Hohmann transfer or prove that all orbital elements match the ideal target.

The associated files are *gmat/hohmann_gto.script* and *results/hohmann_transfer_report.txt*. The text output and script were not directly accessible during report generation; values in this section use the project results reported in the conversation.

6. Discussion

The B-dot simulation demonstrates the expected qualitative behaviour of magnetic detumbling: the angular rate decreases as the magnetic controller applies torque. The gain comparison shows a substantial improvement between gains 10,000 and 50,000, with a smaller improvement at 200,000. The actuator limit is important because higher controller commands cannot exceed the specified maximum dipole moment.

The orbital analysis provides a useful comparison between closed-form two-body calculations and a numerical mission-analysis tool. The reported GMAT apogee is within approximately 0.52 km of the target altitude. For a stronger validation, the initial orbital elements should be documented precisely, the first burn should be applied at the intended perigee, the final orbit's eccentricity and inclination should be checked, and the achieved state should be compared against the full target orbit rather than altitude alone.

7. Limitations

The attitude model uses a simplified Earth magnetic-dipole field and neglects the orbital contribution to the body-frame magnetic-field derivative used by the controller. It assumes ideal measurements and actuators and does not include sensor noise, actuator dynamics, external disturbance torques, or magnetic-torquer imperfections. The inertia tensor is an estimate rather than a verified CAD-derived value. Consequently, the reported detumbling times are preliminary simulation results.

The transfer calculation assumes an ideal two-impulse Hohmann transfer in a two-body model. GMAT output depends on the initial state, coordinate systems, maneuver frame, propagation settings, and event conditions. The close apogee altitude is an encouraging numerical check, but additional orbital-element checks are needed before claiming full transfer validation.

8. Conclusion

A Python model was used to study B-dot attitude detumbling, compare three controller gains, and examine magnetorquer saturation. The nominal case reached below 0.5°/s in 57.91 minutes, while the alternative initial tumble case took 85.47 minutes. An analytical Hohmann transfer from a 500 km orbit to a 35,786 km apogee altitude yielded an ideal first-burn magnitude of 2.3698 km/s, second-burn magnitude of 1.4463 km/s, and transfer time of approximately 5.31 hours. The reported GMAT run used a first-burn value of 2.3672 km/s and reached an apogee altitude of 35,786.52 km. These results illustrate the complementary use of numerical simulation and analytical orbital mechanics, while remaining subject to the model limitations discussed above.

9. References and Tools

[1] Vallado, D. A., *Fundamentals of Astrodynamics and Applications*, for orbital mechanics and two-body transfer relations.

[2] Wertz, J. R. (ed.), *Spacecraft Attitude Determination and Control*, for spacecraft attitude dynamics and magnetic actuation.

[3] NASA / GMAT, *General Mission Analysis Tool*, official software and documentation.

[4] Project implementation files: Python scripts in *src/*, GMAT script in *gmat/*, and numerical outputs in *results/*.

Reproducibility note: Before final submission, verify every numerical value against the saved CSV and GMAT report, and replace the recreated illustrations with the exact saved project plots if required by the evaluator.